Response to the Editor and Reviewer

Submission ID: 3ea58a1a-f4a5-4876-acc2-7c05a67dc433
Original title: Triplet architecture enables deep error-minimization of the genetic code
Revised title: Error robustness of doublet and triplet genetic codes
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To the Editor

We thank the Editor for the opportunity to revise and the Reviewer for a report that was careful and, on every substantive point, correct.

We must state at the outset that the revision reverses the manuscript's central conclusion. In addressing the Reviewer's comments we audited our analysis pipeline and found an error of our own that the Reviewer could not have seen. The Methods described a null model that permutes amino acids among synonymous codon blocks. The code that produced every z-score in the manuscript instead permuted labels among individual codons, which scatters synonymous codons across the codon graph. Under that null the reported z-scores largely measure how tightly synonyms cluster, not how well the assignment minimizes errors: when every amino acid is made equidistant from every other, so that the metric contains no chemistry at all, the same null reproduces the published values ($z = -17.07$ for the standard code against -16.56 reported). Under the block-preserving null that the Methods described, the standard code's assignment remains significantly error-minimizing but modestly so ($p = 0.010$ and 0.027 with the rebuilt descriptors; 0.0005 to 0.027 across all representations), and the doublet-triplet contrast on which the original conclusion rested does not hold.

The question the Editor judged of interest, whether a triplet code is more robust to translation error than a doublet code, is unchanged. The answer is different. We now compare the two architectures directly, by computing the lowest expected cost of misreading per translated residue that any code of each length can achieve, rather than by comparing z-scores against architecture-specific nulls. The revised manuscript shows that:

1. At fixed amino-acid usage, the achievable minimum does not depend on codon length. Without physicochemical distances this follows exactly from the edge-isoperimetric inequality for Hamming graphs (Lindsey 1964); with distances, certified searches give identical doublet and triplet optima under uniform usage and a 1.6% difference under proteome usage.
2. A longer code gains robustness only by making its most error-prone position synonymous. Its advantage is the cost of the channel it discards, which we bound with published position-specific misreading rates. The standard code silences position 3, as in vitro decoding rates predict.
3. The same reasoning explains why codons are not four nucleotides long: the best quadruplet codes we found gain 2 to 6% at measured rates, at the price of a one-third longer message.

Because the conclusion changed, nearly every section was rewritten, and a tracked-changes file would mark almost every line. We therefore provide a clean revised manuscript and this letter, which states what changed and why. The data, descriptors, reduced alphabet, codon projections and Monte Carlo pipeline of the original submission are retained, and every new computation was checked to reproduce the originally reported values to machine precision before it was used. If the Editor considers that the change in conclusion requires a new submission rather than a revision, we will follow that decision.

A list of all corrections to the original Methods follows the point-by-point response.

Point-by-point response

Comment 1. Combinatorial baseline across architectures

> *"A central claim of the paper is that error minimization is an architecture-level property, inferred by comparing z-scores across codes of different sizes. […] the greater statistical significance observed for the triplet code may be, at least in part, a trivial consequence of its larger combinatorial space rather than evidence of a genuine architecture-level effect. I would encourage the authors to address whether their z-score comparison controls for this difference in the size of the null configuration space […]."*

We agree, and the problem was larger than the Reviewer's estimate. The comparison did not control for configuration-space size, and the null used in the code also differed in synonymy structure between architectures (see the opening letter). A z-score measures an observed code against its own ensemble, so z-scores from ensembles of different size and structure cannot be compared.

The revised manuscript states this directly in the section on null models: the codon-shuffling null is a legitimate first choice, it is the null we used in our initial analyses, and it leads to the conclusion of the original submission, which the revised analysis shows to be an artefact of the null. We are grateful to the Reviewer, whose questions about the null configuration space led us to find this.

We have removed every cross-architecture comparison of z-scores. The architecture comparison now rests on the minimum achievable per-residue cost at each codon length (Results, "Codon length does not change the achievable cost"; table 1), which does not depend on any null. Where we still report nulls, we compare the doublet and ten-class triplet projections under the block-preserving null enumerated exhaustively, over configuration spaces of 9! = 362,880 and 10! = 3,628,800 codes, within a factor of ten of each other. On that matched footing the doublet projection ranks in the top 0.5% of its space ($p = 0.0051$) and the ten-class triplet projection is not significant ($p = 0.17$), the opposite of the original conclusion. Because five classes have two members equally close to their centroid, we repeated this for all 32 choices of representatives: the doublet projection is more significant than the ten-class triplet projection in every one (table S1). The doublet projection's significance depends on how the principal components are scaled (table S3), but the ordering of the two projections holds in every representation (Results, "Why codon-shuffling nulls overstate error minimization").

Comment 2. Contribution of PCA

> *"Since the eight principal components (PCs) account for 97.1% of the total variance, the Euclidean distance in PC space would be almost identical to that in the standardized 20-descriptor space, provided the PC scores are not scaled separately. […] the claim that PCA eliminates 'researcher degrees of freedom' is difficult to justify."*

The Reviewer is right about the principle and right to doubt the claim, which we have withdrawn. In checking this comment we found that our pipeline did scale the PC scores separately: each of the eight components was standardized to unit variance before distances were computed, a step the original Methods did not state. This rescaling gives the eighth component (2.1% of the retained variance) the same weight as the first (37%), so it is itself a researcher choice, not a removal of one.

The Reviewer's expectation holds for unscaled components. With the original descriptor matrix, for the standard code against the codon-shuffling null, unscaled components give z = −12.82 (E) and −15.56 (D), and the 20 standardized descriptors give −12.98 and −15.86, against −14.60 and −16.56 with rescaled components. The revised Methods state the rescaling and table S3 reports all main statistics under all three representations, for both the original and the rebuilt matrix. Checking this comment also led us to the provenance of the descriptors themselves. The original matrix had been compiled by hand from published scales, and several values were approximations without a single primary source. We rebuilt it: sixteen descriptors are now AAindex entries cited by accession, four are computed from side-chain structure with RDKit, and two properties with no citable source (side-chain pKa and a strand moment) were omitted (table S2). All physicochemical results in the revision use the rebuilt matrix, and table S3 reports the original matrix and unscaled components as sensitivity analyses; no conclusion changes. The conclusions that matter no longer depend on this choice: the length invariance without chemistry is independent of the distance matrix, and the standard code's significance under the block-preserving null holds under every representation ($p = 0.0005$ to 0.027).

Comment 3. Redundancy between E and D

> *"[…] E = H2 and D ≈ H1. […] the 1/|N(c)| weighting and the normalization constant in D have virtually no effect on the z-score. The very high correlation coefficient (r = 0.96) between the two metrics across the null distribution is thus expected, and the two metrics do not appear to provide 'partially independent evidence'."*

We agree. The implemented D contained no $1/|N(c)|$ factor at all: it equals $2H_1/|C|$ exactly, so D is exactly proportional to H_1, and its correlation with E = H_2 ($r = 0.958$ under the codon-shuffling null, 0.983 to 0.996 under the block-preserving null) is that of H_1 with H_2. The Methods formula was wrong and has been corrected. We have removed the claim of partially independent evidence.

The revision uses a single objective, a usage-weighted and position-weighted generalization of D, because D turns out to be exactly the expected cost of misreading per translated residue, which is the quantity the architecture comparison needs. With equal position weights and usage proportional to codon number it reduces to the original D

(Results, “The per-residue cost of misreading”). E is reported only once, alongside D, for the standard code under the block-preserving null.

Comment 4. Condition C as a control

> *“Condition C is therefore better regarded as a coarse-grained version of Condition B than as an independent control, and the similarity in z-scores between B and C may be expected by construction. It thus remains unclear whether alphabet size genuinely affects the degree of optimization.”*

We agree. Coarse-graining to ten classes raises the number of synonymous edges from 67 to 104 of 263, which the codon-shuffling null rewards whatever the chemistry. Under the block-preserving null, enumerated exhaustively, the ten-class triplet projection is not significant for the representatives used in the main text ($p = 0.17$; 0.13 to 0.67 across descriptor representations), and across all 32 equally valid choices of representatives its p ranges from 0.009 to 0.20, against 0.0001 to 0.017 for the doublet projection: merging chemically close amino acids absorbs the closest neighbour pairs into synonymy and removes much of the assignment signal. We no longer use Condition C as a control. The ten-class alphabet is retained only because a doublet code cannot encode twenty amino acids, and the architecture comparison is made between optimal codes over that alphabet, not between projections of the standard code. The effect of alphabet size is addressed directly by computing the optima for 4, 8, 10, 16 and 20 classes (tables 1 and 3).

Comment 5. Separating architecture from assignment

> *“In all conditions, the null model randomizes only the amino acid assignments while keeping the codon length and degeneracy structure fixed, so the z-score for each condition indicates only how well the assignments are optimized within a given architecture. Comparing these z-scores across conditions does not, by itself, separate the contributions of architecture and assignment to optimization.”*

We agree, and this comment reshaped the paper. To separate architecture from assignment we asked what each architecture permits: the lowest per-residue cost that any assignment can achieve at a given codon length. Two results answer the Reviewer’s question.

First, architecture as such contributes nothing at equal misreading rates. Any doublet code becomes a triplet code of identical cost when a synonymous third position is added (the “silent lift”), so a triplet code can never be worse; and Lindsey’s edge-isoperimetric inequality shows that without chemistry it can never be better, since the minimum number of non-synonymous errors per residue, $3 \log_4 k$ for k equally used classes, does not depend on codon length. Certified searches confirm equality with chemistry (21.345 for both lengths, ten classes, uniform usage) and for twenty amino acids between triplets and quadruplets (6.80 without chemistry; identical with chemistry at uniform usage).

Second, architecture contributes only through position-specific misreading rates: a longer code can make its noisiest position synonymous. The resulting advantage is 0 to 32% across published rate sets and assumptions about doublet decoding (table 2), and the standard code’s synonymy pattern matches in vitro decoding rates. We state plainly in the

revision that the size of the doublet-triplet difference depends on how accurately a doublet decoder would read its last base, which cannot be measured.

Comment 6 (minor). Figure 1a

> *"[…] the edges intersect in a complex manner, making it difficult to discern the overall graph structure of the code. It may be helpful to consult graph-representation methods used in other studies (e.g., 10.3390/sym7031211)."*

We thank the Reviewer for the reference (Lenstra 2015, Symmetry 7:1211-1260). The former Figure 1a has been replaced. The new Figure 1 draws the Hamming graphs $H(2, 4)$ and $H(3, 4)$ position by position, following the Reviewer's suggestion and citing Lenstra (2015), and illustrates the silent lift that the revised argument depends on: each doublet codon becomes a column of four triplet codons sharing its class, so that position 3 is synonymous and each codon keeps its doublet neighbours.

Corrections to the original Methods identified by the authors

In addition to the points raised by the Reviewer, the audit found the following. Each is corrected or removed in the revision.

1. **Null model.** The Methods and the Figure 2 caption described permutation among synonymous blocks with AUG fixed to Met. The code permuted labels among all 61 sense codons, preserving codon counts per amino acid, and did not fix AUG. The revision reports both nulls under their correct descriptions.
2. **Principal-component rescaling.** Each component was standardized before distances were computed; now stated (Comment 2).
3. **Definition of D.** The printed formula contained a $1/|N(c)|$ factor absent from the computation; corrected (Comment 3).
4. **Doublet projection.** The majority-vote doublet projection contains nine classes, not ten: His loses its only prefix (CA) to the Asn/Gln class in a tie, which was broken by the order of the class identifiers (X1 to X10); there are four such ties (AA, AG, CA, UU). A projection that keeps all ten classes exists only if each tie is broken toward the class that would otherwise lose its last prefix; it is reported in table S1 (block-preserving $p = 0.043$, against 0.17 for the ten-class triplet projection). The revision states this and treats the projection as a nine-class code.
5. **Representative amino acids.** For the five two-member classes both members are equidistant from the class centroid, so the "closest to centroid" rule does not choose between them; the choice was made by floating-point rounding. The revision states that the choice is arbitrary and reports alternatives in table S1.
6. **Best random doublet.** The abstract reported the best random doublet code at $z = -4.0$. In our replication of that null with the original descriptor matrix (10^6 draws), the minimum was $z = -5.00$ (D) and -4.90 (E); a sample minimum depends on the draw, and an independent run gave -4.76 and -4.37, so the reported -4.0 understated it in either case. The statement has been removed with the analysis it belonged to.

7. **Position-specific synonymy.** The original reported 69 to 77% synonymy at position 3 and 10% at position 2. Counting single-nucleotide substitutions between sense codons, the values are 71.6% at position 3, 4.6% at position 1 and 0 at position 2. The revision uses these values and states the definition.
8. **Comparison with Freeland and Hurst (1998).** The claim that our result strengthened theirs by two orders of magnitude rested on the codon-shuffling null. Under the block-preserving null their comparison uses, our p-values (0.010 and 0.027 with the rebuilt descriptors) are weaker than theirs. The claim has been withdrawn.
9. **Descriptor provenance.** The Methods stated that the descriptors were taken from AAindex. They were compiled by hand, with a correlation filter at $|r| \geq 0.95$ that the Methods did not state, and several values were approximations without a primary source. The matrix has been rebuilt from AAindex accessions and structural rules (Comment 2), and the filter and its sensitivity are stated.

Data and code

All scripts, verification checks and outputs are archived in a new version of the Zenodo record accompanying the original submission (https://doi.org/10.5281/zenodo.21614179, concept DOI, which resolves to the latest version) [add the new version DOI once published]. Every revised result was computed only after the corresponding original values had been reproduced to machine precision from the archived data, and the archive records these checks.

We are grateful to the Reviewer. The revised manuscript answers the question the original set out to answer, and it does so because the Reviewer asked the right questions of the method.

Sincerely,

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